

Tap-to-Adapt: Learning User-Aligned Response Timing for Speech Agents

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Abstract

Response timing judgment is a critical component of interactive speech agents. Although there exists substantial prior work on turn modeling and voice wake-up, there is a lack of research on response timing judgments continuously aligned with user intent. To address this, we propose the Tap-to-Adapt framework, which enables users to naturally activate or interrupt the agent via tap interactions to construct online learning labels for response timing models. Under this framework, Dilated TCN and a sequential replay strategy play significant roles, as demonstrated through data-driven experiments and user studies. Additionally, we develop an evaluation and continuous data mining system tailored for the Tap-to-Adapt framework, through which we have collected approximately 20,000 samples from the user studies involving 20 participants.

1 Introduction

With the emergence of reasoning capabilities in large language models (LLMs), interactive speech agents enhanced by LLMs have demonstrated increasingly remarkable performance, being widely applied in mobile devices [Deng et al., 2024, Liu et al., 2025b], wearable devices [Mishra et al., 2025, Ren et al., 2025], smart home systems [Rivkin et al., 2024, Lin et al., 2025], in-vehicle systems [Furusawa and Saitoh, 2024, Furusawa, 2025], elderly care [Yang et al., 2024, Liu et al., 2025a], etc. Determining whether and when to respond to a user is a critical component of interactive speech agent systems [Skantze, 2021, Walker, 2024, Liu et al.,

2025a, Ethiraj et al., 2025].

One line of relevant work focuses on modeling the start timing of utterances in speech dialogue scenarios; however, these methods only reflect the objective turn-initialization in conversations and do not capture the timing at which a human expects the agent to respond [Skantze, 2017, Ekstedt and Skantze, 2022, Li et al., 2022, Wagner et al., 2025]. Another line of work concerns voice wake-up tasks, which mostly rely on keyword spotting or speaker recognition mechanisms, and cannot perform response timing judgments or cessation in multi-turn interactions aligned with user intent [Kępuska and Klein, 2009, Kępuska, 2011, Kępuska and Bohouta, 2017], exhibiting a significant mismatch with human-speech agent interaction. Currently, there is a lack of work that drives human-agent interaction by aligning response timing with human intent in multi-turn dialogue contexts.

On the other hand, user tap interactions on screens, accompanied by gesture and facial information, help systems recognize specific users [Choi and Park, 2015, Wang et al., 2021, Khamis et al., 2022], indicating the feasibility of constructing personalized labels through target-user tap behavior.

Motivated by these observations, we propose the Tap-to-Adapt framework for interactive speech agents, which allows users to manually activate or interrupt agent responses through simple tap actions, constructing "positive" (from activation) and "negative" (from interruption) online learning labels. Here, *tap* serves as an abstract interaction prototype representing explicit user intervention, extensible to other interaction modalities in practice. Within this framework, we find that combining a Dilated TCN

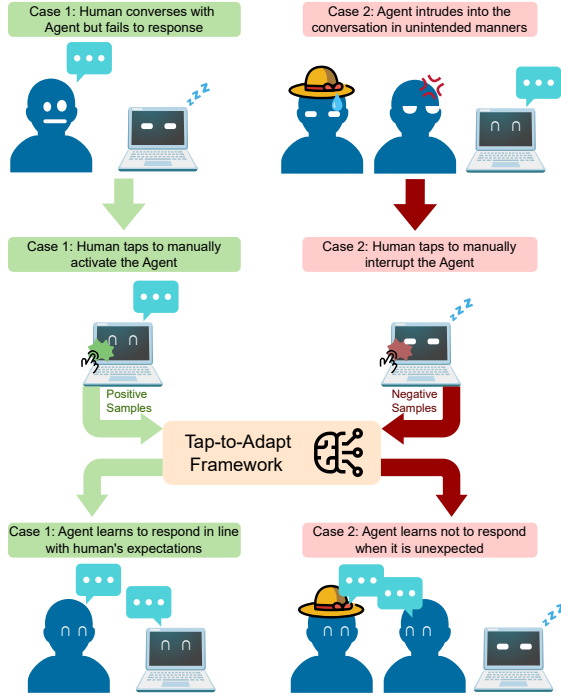


Figure 1: In human–speech agent interaction scenarios, the Tap-to-Adapt Framework drives the agent to learn to respond according to the human’s expectations by constructing online learning samples through manual activations and interruptions performed by the human.

encoder [Lea et al., 2017] with a label-balanced sequential replay online learning strategy [Rolnick et al., 2019, Ji et al., 2021, Liu et al., 2024] yields significant performance gains. The model leverages online learning labels to learn response timing aligned with user intent from speech context.

Additionally, we propose a unified system for evaluating the Tap-to-Adapt framework and constructing response-timing judgment data in human–agent interaction scenarios, driven by both controlled simulation and user studies. Using this system, we simulate 1,000 temporally sequenced multi-scenario speech samples and conduct user studies with 20 participants, resulting in 2,478 online learning samples labeled from interaction behaviors, as well as a large-scale temporally sequenced dataset collected from real interactions. From the latter, we extract 17,416 multi-user speech samples. These data help mitigate the lack of evaluation benchmarks for response timing aligned with user intent in human–agent interaction scenarios. Leveraging this system and dataset, we further validate the effectiveness of the model and learning strategy.

In summary, our contributions are:

1. We propose **Tap-to-Adapt**, a framework that constructs online learning supervision signals through natural user tap actions, enabling the speech model to proactively align with user-specific expected response timing in real-time speech agent interactions. This framework addresses the challenge of aligning agent speech response timing with user intent.
2. Within this framework, we find that a Dilated TCN-based model combined with a sequential sample replay online learning strategy demonstrates clear effectiveness. The approach enables efficient adaptation under cold-start, pre-training, user-specific weights, and multi-user shared weights scenarios, without relying on offline manually constructed samples.
3. We develop a systematic system for evaluating the Tap-to-Adapt framework and constructing response-timing data, combining controlled simulation with user studies. Using this system, we mined approximately 20,000 samples, helping fill the gap in datasets for response timing aligned with user intent in human–agent interaction scenarios.

The source code is available at https://github.com/ZihongHe/tap_to_adapt.

2 Related Work

2.1 Speech Turn-Initialization Modeling

Our work aims to learn a turn-initiation speech model aligned with user preferences to identify the appropriate timing for a speech agent to respond. Speech turn-initiation modeling can be considered a subset of turn-taking modeling, referring to the prediction of when speech activity begins [Skantze, 2017, Ekstedt and Skantze, 2022, Li et al., 2022, Wagner et al., 2025]. There are several representative works. Skantze et al. [Skantze, 2017] used LSTM-RNNs to estimate the probability of a speaker initiating speech in a short future window for each frame and evaluated the model by predicting the speaker within 1 second after valid pauses. Ekstedt et al. [Ekstedt and Skantze, 2022] further employed CPC- and Transformer-based models to capture temporal dependencies within future speech activity windows and validated the model on turn-switching detection in two-speaker audio dialogues. Addressing the limitations of pause-based triggers

in turn-initiation modeling, which differ from natural human dialogue with rapid turn-taking and overlaps, Li et al. [Li et al., 2022] used Gaussian Mixture Models (GMMs) to predict continuous timing for dialogue initiation and validated their approach on speech dialogue data. However, current turn-initiation models only align with the actual timing of speaker utterances in real speech dialogues. These works exhibit a gap in human–speech agent interaction, as they cannot appropriately model the timing when a user expects the agent to respond.

2.2 Speech Wake-Up Systems

Speech wake-up refers to the task of activating a system via specific vocal signals [Kěpuska and Klein, 2009, Kěpuska, 2011, Kěpuska and Bohouta, 2017]. Broadly, our Tap-to-Adapt Framework falls under the category of speech wake-up systems. These systems are widely applied in wearable devices [Schmalfuß-Schwarz et al., 2024], home voice assistants [Ferro Filho et al., 2024], voice assistants for elderly care [Yan et al., 2024], etc. Early speech wake-up systems were mostly based on Hidden Markov Models (HMMs), Gaussian Mixture Models (GMMs), or hybrid GMM-HMM approaches. While these methods adapt well with limited data, they exhibit lower robustness to speaker variability and noise [Vroomen and Normandin, 1992, Ghaffarzadegan et al., 2017, Schröder et al., 2016]. Recently, neural network–based solutions have shown higher reliability [Oord et al., 2016, Shan et al., 2018, Li et al., 2019, Gulati et al., 2020, Jose et al., 2020, Radford et al., 2023]. For example, Jose et al. [Jose et al., 2020] used convolutional neural networks (CNNs) for wake-word detection with precise start–end localization, and Sahai et al. [Sahai et al., 2023] proposed a dual-attention wake-word detection model, one focusing on the wake-word and another on broader speech context, to improve accuracy. However, most speech wake-up systems still rely on fixed wake-words [Kěpuska and Klein, 2009, Kěpuska, 2011, Kěpuska and Bohouta, 2017, Jose et al., 2020], and few provide mechanisms to align with real user interaction labels [Mallidi et al., 2018, Norouzian et al., 2019, Dighe et al., 2024], limiting their ability to match actual user intent.

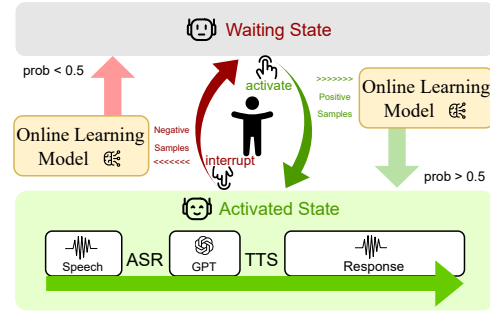


Figure 2: The figure illustrates the architecture of the Tap-to-Adapt Framework.

2.3 User-Feedback-Based Online Speech Learning

Our system learns in real time from speech data generated by user screen taps during interactions. There are several representative works in user-feedback-based online speech learning. Gavsic et al. [Gašić et al., 2011] allowed users to provide binary satisfaction ratings in dialogue systems to optimize response strategies via online learning. Su et al. [Su et al., 2015] mined naturally occurring interaction behavior during dialogues to optimize models through online reinforcement learning, improving performance, while Su et al. [Su et al., 2016] incorporated subjective feedback on dialogue success as online RL rewards. Zhou et al. [Zhou et al., 2023] proposed a speech-to-text system in which users manually corrected recognition errors to optimize the model, and He et al. [He et al., 2025] enabled users to provide real-time verbal feedback during speech recognition and speaker separation to correct errors, e.g., identifying the correct speaker. However, these approaches either rely on deliberate user feedback [Gašić et al., 2011, Su et al., 2016, Zhou et al., 2023, He et al., 2025] or on implicit signals arising during interaction [Su et al., 2015]. Currently, there is a lack of studies exploring explicit and naturally occurring user feedback signals in online speech learning tasks.

3 Method

3.1 Framework Overview

Following the continuous prediction approach in [Inoue et al., 2024, Castillo-López et al., 2025], the Tap-to-Adapt framework performs continuous predictions using the latest 15 seconds of audio as in-

put with a 0.2-second time interval. The specific workflow is illustrated in Figure 2. When the on-line learning model predicts a probability below 0.5, the agent remains in a silent state; when the prediction exceeds 0.5, it activates the agent to enter the *Activated* state and ultimately complete the response. While in the *Waiting* state, a user tap forcibly drives the agent into the *Activated* state, and the audio context at that moment is used to construct a positive online learning sample for the model. Conversely, while in the *Activated* state, a user tap forcibly returns the agent to the *Waiting* state, and the corresponding audio context is used to construct a negative online learning sample for the model.

3.2 Model and Learning Strategy

Dilated TCN-based Model In our response prediction based on human–LLM agent voice interactions, both large-scale and small-scale temporal dependencies play a critical role. Large-scale dependencies include the sequential order of the dialogue, such as whether the agent speaks before the human, while small-scale dependencies involve the content of utterances, e.g., determining whether a dialogue should end based on the final words or intonation of the human speaker. This indicates that the model needs to capture multi-scale temporal dependencies. Dilated TCNs are more efficient than standard TCNs in capturing long-range speech dependencies [Lea et al., 2017]. Combined with experimental validation, Dilated TCN-based model demonstrates excellent performance as the online learning model within the tap-to-adapt framework. Fig. 3 illustrates the model we constructed for this framework. The model input is a Log-Mel Spectrogram extracted from raw audio sampled at 16 kHz and downsampled to 4 kHz [Davis and Mermelstein, 1980]. We further normalize the audio input using Z-score normalization to eliminate amplitude variations, allowing the model to focus on key features rather than loudness [Fawcett et al., 2021].

Sequential Replay Online Learning Strategy

As for the online learning method, we draw inspiration from prior work [Rolnick et al., 2019, Ji et al., 2021, Liu et al., 2024] and adopt a sequential replay online learning strategy, with label balancing as the default configuration. Specifically, at each interaction step t , the model first generates a prediction for the incoming sample (x_t, y_t) following a strict

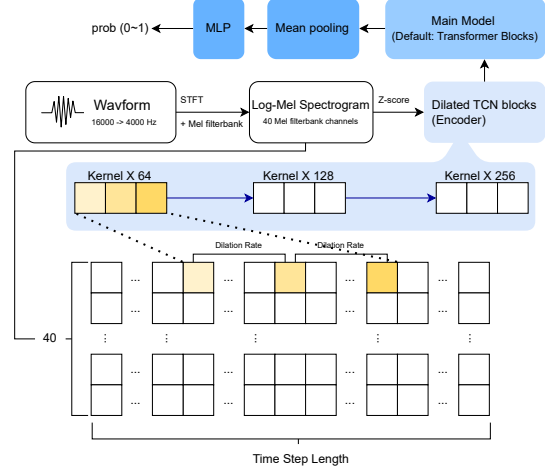


Figure 3: The figure illustrates the architecture of an efficient **Dilated TCN-based model** within the **Tap-to-Adapt** framework. The model first extracts Mel-spectrogram features from the input audio, then passes them through multiple TCN blocks with a dilation mechanism to enlarge the temporal receptive field. The resulting features are subsequently fed into a main model, which by default adopts Transformer blocks [Vaswani et al., 2017], consisting of an Input Projection Layer, Positional Encoding, and a Transformer Encoder.

predict-then-update protocol. After evaluation, the model performs a series of sequential replay updates on a mini-batch $\mathcal{S}_t$ consisting of several recent samples stored in a replay buffer. To emphasize recent user behavior, each sample $(x_i, y_i) \in \mathcal{S}_t$ is assigned a weight that decays exponentially with its age:

$$w_i = w_0 \cdot \gamma^{t-i},$$

where w_0 is the original sample weight and $\gamma \in (0, 1)$ is the sliding-window decay factor. This weighted replay ensures that more recent interactions have greater influence during updates while older samples still contribute, effectively mitigating catastrophic forgetting.

3.3 Sample Mining and Evaluation

Simulated Sample Construction The Tap-to-Adapt framework supports the simulation of human–speech agent interaction scenarios through predefined, probability-driven scene configurations. The framework supports background sound simulation using the DataSec dataset, which contains 22 top-level sound categories [Fredianelli et al., 2025]. For each category, the longest audio sample can be selected and played with volume scaling within a



Figure 4: Results of the validation experiment using 1,000 simulated samples for Tap-to-Adapt Framework. The related model and online learning strategy are described in Sec. 3.2. Here, without Transformer Blocks indicates that the main model is implemented as a simple MLP.

predefined range to emulate varying background sound levels in real-world environments. After playback, a silent segment of equal duration is inserted, after which another audio sample is randomly selected and the process repeats. Dialogue topics follow the configuration described in [Li et al., 2017]. To preserve scene continuity, the framework allows certain scenario-related factors to be assigned recurrence probability weights so that they can persist across consecutive scenarios. These factors are summarized in Appendix A. Positive samples occur only when the user is speaking to the agent and the sample type corresponds to “finished speaking, expecting a reply.”

Real Sample Mining The Tap-to-Adapt framework supports real sample mining during actual use by leveraging user interactive behaviors and state information. Specifically, it collects: (1) The audio input corresponding to the most recent user manual activation of the Tap-to-Adapt model as a positive sample; (2) The audio input at the start of the Activated State when the user interrupts as a negative sample; (3) The audio input at the moment when the model prediction exceeds 0.5 and the user does not interrupt as a positive sample; (4) Additional negative samples are randomly selected from all samples with model prediction below 0.5, excluding the 20 samples immediately prior to a user manual activation, to expand the negative sample set.

4 Experiment

4.1 Validation Experiment

Experimental Method: The speech agent system was run on an x64-based computer with touchscreen support to provide the interface. Following [Coucke et al., 2019], “Snips” was used as the agent’s designation, but it was not necessary to trigger the system with the designation at the beginning of each interaction. Based on Simulated Sample Construction in Sec. 3.3, 1000 sequentially ordered simulated data samples were constructed and serialized learning was conducted along the temporal order of the data, performing a prediction before learning each sample to simulate real interaction scenarios. We investigated the effectiveness of the Dilated TCN-based model and sequential replay online learning strategy (Sec. 3.2) in the task of user-aligned speech response timing prediction through ablation studies, and further compared the Dilated TCN-based model with representative existing models, Conformer [Gulati et al., 2020] and Jasper [Li et al., 2019].

Evaluation Metrics: Every 100 samples, the F1 score and Brier Score were computed, where the Brier Score measures the mean squared error between predicted probabilities and actual outcomes [Glenn et al., 1950].

Evaluation Result: Figure 4 presents the experimental results evaluating the effectiveness of the model and learning strategy. It can be observed that the introduction of the dilated TCN and replay-based learning strategies yield the most significant performance improvement. Moreover, the Dilated TCN-based model demonstrates overall superior

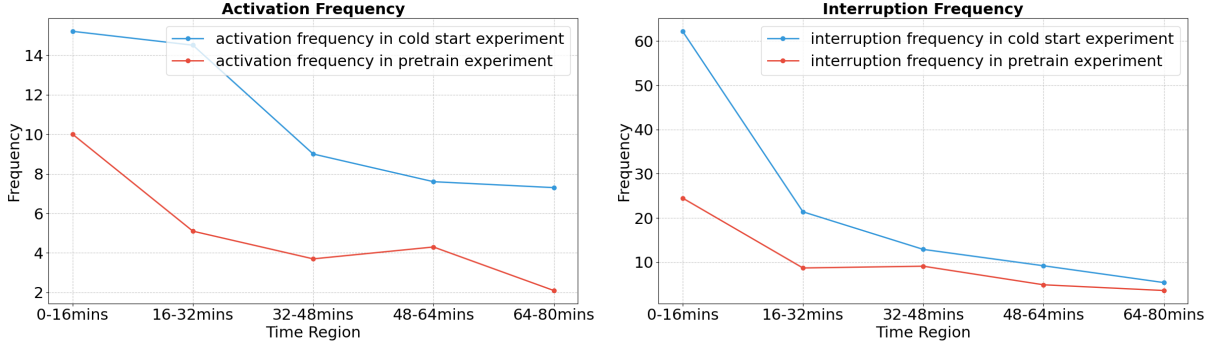


Figure 5: The temporal trend of average activation and interruption frequencies in continuous participant interactions under consistent environmental conditions over time.

Setting	Trend-Oriented Questionnaire							Overall Questionnaire						
	Q1	Q2	Q3	Q4	Q5	Q6	Avg	Q1	Q2	Q3	Q4	Q5	Q6	Avg
Cold-start	5.9	5.9	6.4	6.0	6.2	6.2	6.1	5.4	5.5	5.1	6.2	5.6	5.6	5.6
Pre-trained	5.9	5.5	5.4	5.8	6.3	5.9	5.8	5.6	5.3	5.2	6.2	5.8	5.8	5.7

Table 1: User study rating results for trend-oriented and overall questionnaires. Q1–Q6 correspond to System Response Accuracy, Likeability, Cognitive Demand, Annoyance, Habitability, and Speed, respectively.

performance to Conformer and Jasper. In contrast, the use of Transformer blocks as the internal configuration of the main model and the label balance mechanism in the learning strategy show limited impact on performance.

4.2 Usability Experiment

Experimental Setup: The experimental setup follows the same hardware configuration as Section 4.1. We informed the participants of the data usage and potential risks, and obtained their signed consent forms. We recruited 20 participants (gender-balanced, 1:1), who were randomly divided into two groups. The first group conducted an online learning experiment under a cold-start setting (with the untrained initial model), while the second group performed online learning based on a pre-trained model constructed from the activation and interruption samples of the first group.

Participants were aged between 18 and 30 years old, a demographic shown to have relatively high acceptance of intelligent agents [Ho et al., 2022, Hail et al., 2024], with a mean age of 24. The experiment for each participant lasted 80 minutes. Throughout the session, participants continuously interacted with the agent; however, every 6 minutes, the participant engaged in a 2-minute casual conversation with the experimenter. This alternating design taught the agent to avoid inappropriate interventions during background conversations. At

the same time, these 2-minute conversations served as periodic micro-breaks that helped alleviate participant fatigue. When the agent responded late, prematurely, or at unsuitable moments, users manually corrected it by tapping the screen to activate or interrupt the agent; meanwhile, the online learning mechanism continuously aligned the agent’s response timing with the user’s intent using samples constructed from these tapping-based labels. Background music was played within a specific volume range, following the setup described in Sec. 3.3 on Simulated Sample Construction. Participants could choose dialogue topics from [Li et al., 2017] to initiate conversations.

After the experiment, questionnaires were divided into two types: (i) a *trend-oriented questionnaire* (detailed in Appendix B.1, capturing participants’ perceived improvement of the system over time, and (ii) an *overall questionnaire* (detailed in Appendix B.2, reflecting participants’ holistic impressions of the system. Both questionnaires were designed based on the SASSI framework [Hone and Graham, 2000], covering six dimensions with six questions each. All items were rated on a 7-point Likert scale (1: strongly disagree, 7: strongly agree).

Evaluation Metrics: The questionnaires assessed six dimensions derived from SASSI [Hone and Graham, 2000]: (1) System Response Accuracy, (2) Likeability, (3) Cognitive Demand, (4) Annoyance, (5) Habitability, and (6) Speed.

After completing the questionnaires, semi-structured interviews were conducted, including questions about participants’ overall impressions of the agent system as well as the motivations behind notable questionnaire ratings.

Evaluation Results: Experimental results are presented in Figure 5. The figure shows that pre-trained models reduced participant activation and interruption frequencies compared to the cold-start setting, with both settings gradually converging to a low interaction frequency and demonstrating effective adaptation. Further statistical analysis confirmed that both activation and interruption frequencies changed significantly over time (Friedman test, $p < 0.01$) and significantly decreased from the first to the last interaction period across both settings (Wilcoxon signed-rank tests, $p < 0.01$).

The average questionnaire scores are presented in Table 1. Further statistical analysis using one-sample t-tests verified that both the trend-oriented and overall evaluation scores for both groups were significantly higher than the neutral midpoint of 4 ($p < 0.001$). Furthermore, a Wilcoxon signed-rank test within the cold-start group revealed that their trend-oriented scores were significantly higher than their overall evaluation scores ($p = 0.0078$).

Analysis of experimental observations and interview transcripts reveals several findings:

Finding 1 (Mutual Adaptation). While participants adapted the system through activation and interruption signals, they also adjusted their speaking rate and pause duration to better accommodate the agent, jointly improving conversational fluency.

Finding 2 (Acceptance-Driven Personalization). Participants exhibited varying tolerance to the agent’s early or delayed responses. Some participants would stop speaking to wait for the agent when it responded prematurely, while others would patiently wait for a delayed response after finishing their utterances. These differences in tolerance led to individualized variations in the model’s fitted results. Moreover, we found that models for participants with higher tolerance adapted to their preferred response patterns more rapidly.

Finding 3 (Cultivation Effect). Participants reported a strong perception of progressive improvement, especially in the cold-start group. Some participants described feelings of achievement, trainability, and increased intimacy with the agent, leading them to perceive the system as more distinctive compared to conventional voice assistants.

Finding 4 (Formation of synchrony). Over time, the model’s predictions exceeding 0.5 (which triggered agent activation) increasingly aligned with participant tap timing on the screen, indicating that the expected response times of participants and the agent progressively converged.

We obtained 2,478 online learning samples with ground-truth labels derived from participants’ activation and interruption behaviors. We further expanded the dataset by applying the method outlined in Real Sample Mining (Sec. 3.3) to the massive interaction audio data collected during the usability experiment. This process yielded 17,416 labeled audio samples.

4.3 Extended Experiment

Based on the data accumulated from the user study, we conducted extended experiments covering two aspects. First, we used the 17,416 mined samples to reproduce the significant conclusions obtained in Sec. 4.1 and evaluate performance under user-specific and user-shared weight settings. Second, we investigated whether our method implicitly learned semantic information from the audio signals to determine if it can effectively achieve language comprehension without relying on auxiliary language conversion or enhancement modules.

Performance on Mined Samples We conducted experiments with the following settings:

Setting 1, User-specific weight training: For each participant, their corresponding audio samples were trained sequentially in chronological order. Before each sample was trained, a prediction was first generated to compute evaluation metrics, simulating a realistic cold-start online learning scenario. Final results were reported as averages across participants.

Setting 2, User-shared weight training: All mined audio samples were merged under the constraint that the temporal order of samples from each individual participant was preserved before and after merging. Samples were then randomly interleaved across participants and used for serialized training. The evaluation protocol followed that of Setting 1, simulating an online learning scenario with shared model weights across multiple users.

The experimental results are illustrated in Figure 6. To make the comparison more focused, we retained only the experimental groups from the Validation Experiment that showed relatively clear performance gaps with respect to the Dilated TCN-based

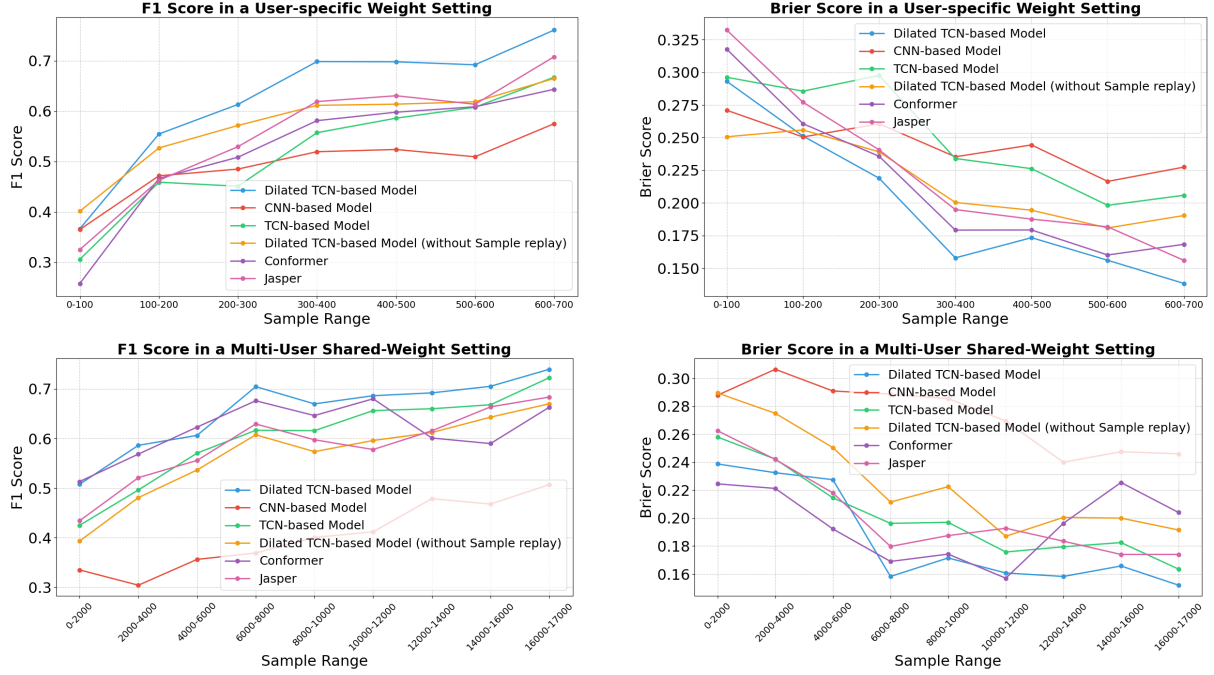


Figure 6: Experimental results in Section 4.3 use the subset of groups selected from the Validation Experiment that produced relatively pronounced differences against the Dilated TCN-based Model. The top row reports the **F1 score** and **Brier Score** under **User-specific Weight Training**, while the bottom row reports the corresponding **F1 score** and **Brier Score** under **User-shared Weight Training**. This focused comparison highlights the effectiveness of the Dilated TCN-based model and the Sequential Replay Online Learning Strategy.

model. The results indicate that the mined data remain learnable under both user-specific weight training and user-shared weight training settings. Moreover, this confirms the effectiveness of the model and learning strategy described in Sec. 3.2.

Semantic Learning Analysis We conducted extra experiments, and the results demonstrate that our proposed method can effectively learn language semantic information from audio signals. We transcribed responses from 20 participants using ASR and encoded sentences via *paraphrase-multilingual-MiniLM-L12-v2*¹. Model predictions and ideal positives formed semantic subspaces. For a 10-step time window t , average pairwise cosine similarities $C_{\text{pred}}(t)$ and $C_{\text{true}}(t)$ were computed, with discrete growth rates:

$$\Delta C_{\text{pred}}(t) = C_{\text{pred}}(t) - C_{\text{pred}}(t-1), \quad (1)$$

$$\Delta C_{\text{true}}(t) = C_{\text{true}}(t) - C_{\text{true}}(t-1). \quad (2)$$

Then, the growth difference between the two tracks was derived as $D(t) = \Delta C_{\text{pred}}(t) - \Delta C_{\text{true}}(t)$.

The cumulative sum of the $D(t)$ sequences was computed for all windows, and the trend slope was

¹<https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>

estimated using linear regression, defined as the Semantic Learning Index (SLI). Significance of the SLI was assessed via bootstrap resampling to obtain the 95% confidence interval (CI). The results are: $\text{SLI} = 0.0116$, $95\% \text{ CI} = [0.0088, 0.0148]$. This confirms that our method develops stable semantic structures beyond the ideal subspace.

5 Future Work

Future research on the Tap-to-Adapt framework could focus on the following directions.

- **User-specific labels:** combining taps with gestures and facial cues to ensure labels originate from the target user;
- **Continuous adaptation:** continuously adapting to users' evolving interaction habits, environments, and devices;
- **Intent augmentation:** constructing user intent labels from signals beyond taps, e.g., facial expressions or semantic cues;
- **Adaptive stopping:** exploring adaptive speech stopping timing;
- **Visual input:** incorporating visual signals to increase the richness of signal sources;

- **Multi-stage systems:** employing edge–cloud collaboration to improve efficiency;
- **Task generalization:** extending the proposed Tap-to-Adapt framework to new tasks, e.g., generative dialogue.

On the other hand, this framework must be used appropriately to prevent risks such as the disclosure of user preferences or the inappropriate collection of user interaction audio.

6 Conclusion

To address the current lack of research on modeling response timing for interactive speech agents in alignment with user intent, we propose Tap-to-Adapt, an online learning framework that learns response timing through users’ natural tap-based activation and interruption behaviors. Within this framework, we find that a Dilated TCN-based model combined with a Sequential Replay Online Learning Strategy achieves significant performance gains. Furthermore, we propose a Simulated Sample Construction and real sample mining system tailored to this task. Using this framework, we conduct user studies as well as ablation and comparison experiments on approximately 20,000 samples to validate both the feasibility of the task and the effectiveness of our approach. Our framework lays the foundation for task exploration, optimization of models and training mechanisms, and dataset expansion.

Limitations

In Section 4, we validated the feasibility of the proposed Tap-to-Adapt framework, as well as its associated models and training strategies, by combining data-driven ablation and comparative experiments with user studies. However, the current sample scale is still insufficient to support more fine-grained conclusions, such as the upper performance limits achievable through pre-training for response timing prediction models, or their fitting behavior under continuously evolving interaction scenarios.

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A Scenario Factors for Simulated Sample Construction

Positive samples are defined when the user is speaking to the agent and the sample type is “finished speaking, expecting reply”. The factors for simulation are described as follows:

- **Dialogue topic with the agent:** “Ordinary Life”, “School Life”, “Culture & Education”, “Attitude & Emotion”, “Relationship”, “Tourism”, “Health”, “Work”, “Politics”, or “Finance”. The probability of repeating the previous topic is set to 50%.
- **Distance to the agent:** “30cm”, “60cm”, or “90cm”. The probability of repeating the previous distance is 50%.
- **Audio containing partial dialogue context:** “Yes” or “No”.
- **User speaking target:** “Agent” or “Other nearby person”. The probability of repeating the previous target is 50%.
- **Sample type:** “Finished speaking, expecting reply”, “Finished speaking, not expecting reply”, “Speaking, pausing”, “Speaking, not pausing”, or “Not speaking”. The type “Finished speaking, expecting reply” occurs four times more frequently than any other type.

B Questionnaire

B.1 Trend-Oriented Questionnaire

1. **System Response Accuracy**
As the number of interactions increases, do you feel that the system’s response timing increasingly aligns with your expectations?
2. **Likeability**
As the number of interactions increases, do you find your favorability toward the system gradually improving, and do you increasingly consider it worth continuing to use?

3. **Cognitive Demand**

As the number of interactions increases, do you feel that the attention and mental effort required during interaction decrease, making the process easier and less demanding?

4. **Annoyance**

As the number of interactions increases, do you feel that interacting with the system becomes less frustrating and produces less sense of failure?

5. **Habitability**

As the number of interactions increases, do you gradually understand more clearly what to say to the system, anticipate its response timing, and feel less confused during interaction?

6. **Speed**

As the number of interactions increases, do you feel that interacting with the system becomes increasingly efficient?

B.2 Overall Questionnaire

1. **System Response Accuracy**

To what extent do you agree that the system’s response timing meets your expectations?

2. **Likeability**

How favorable is your impression of the system, and to what extent do you consider it worth continuing to use?

3. **Cognitive Demand**

To what extent do you agree that the attention and mental effort required during interaction are minimal, making the process relatively easy and effortless?

4. **Annoyance**

To what extent do you agree that interacting with the system is not frustrating and produces minimal sense of failure?

5. **Habitability**

To what extent do you clearly understand what to say to the system, anticipate its response timing, and feel less confused during interaction?

6. **Speed**

To what extent do you agree that interacting with the system is efficient?